PROJECT REPORT
ON

CyberShield-AI: AI-Powered Cyber Threat Intelligence, Detection, Analysis and Attack Prediction Platform

Submitted by

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In fulfillment for the award of the degree
Of
BACHELOR OF TECHNOLOGY
In
Information Technology



INSTITUTE OF TECHNOLOGY AND ENGINEERING
INDUS UNIVERSITY CAMPUS, RANCHARDA, VIA-THALTEJ
AHMEDABAD-382115, GUJARAT, INDIA
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September-2026

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ON

CyberShield-AI: Civic & Enterprise Cyber Threat Intelligence, Detection, Analysis and Attack Prediction Platform

AT

ज्ञानेन प्रकाशते जगत्
INDUS UNIVERSITY

In the fulfilment of the requirement
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PREPARED BY

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SUBMITTED TO
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CANDIDATE'S DECLARATION

We declare that final semester report entitled "CyberShield-AI: AI-Powered Cyber Threat Intelligence, Detection, Analysis and Attack Prediction Platform" is our own work conducted under the supervision of the guide Ms. Sheatal Panchal.

We further declare that to the best of my knowledge, the report for B. Tech final semester does not contain part of the work which has been submitted for the award of B. Tech Degree either in this university or any other university without proper citation.

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CERTIFICATE

Date: 30/09/2026

This is to certify that the project work entitled "CyberShield-AI: AI-Powered Cyber Threat Intelligence, Detection, Analysis and Attack Prediction Platform" has been carried out by Yash Raj (IU2341220107), Ganesh Kondekar (IU2341220102), Jeet Borse (IU231220139), Bhavya Vaidya (IU2341220099), Shiven Asopa (IU2341220142) under my guidance in partial fulfillment of degree of Bachelor of Technology in Information Technology (7th semester of Indus University, Ahmedabad during the academic year 2026 – 2027).

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ABSTRACT

CyberShield-AI is an artificial intelligence-driven cyber threat intelligence, anomaly detection, and predictive attack analysis platform engineered to protect modern enterprise digital assets and critical infrastructure. Contemporary Security Operations Centers (SOCs) face acute operational challenges: security teams are inundated by millions of heterogeneous network events and uncoordinated alerts daily, manual log correlation causes severe analyst fatigue, conventional SIEM platforms rely on brittle static signatures that fail to identify zero-day threats, and disparate telemetry feeds lack unified contextual intelligence. CyberShield-AI resolves these shortcomings with an integrated, high-throughput intelligence ecosystem that enables security engineers and SOC analysts to ingest raw telemetry, isolate anomalous vectors, investigate root causes, and initiate incident containment in under 60 seconds.

The platform serves three dedicated operational roles: SOC Analysts (Triage and Incident Responders), Threat Intelligence Specialists, and Central System Administrators. Analysts inspect incoming event streams, review anomaly scores, filter telemetry across severity levels, and visualize global threat origins across an interactive Leaflet and OpenStreetMap geospatial attack map. High-throughput ingestion pipelines normalize multi-source security telemetry (including firewall logs, authentication attempts, DNS queries, and network flows), while automated IP reputation scoring and geographic coordinate resolution pinpoint attack vectors in real time, filtering benign traffic and prioritizing critical enterprise vulnerabilities.

A dedicated Python FastAPI machine learning microservice delivers real-time unsupervised anomaly detection and attack forecasting by orchestrating an ensemble of Isolation Forest (200 isolation trees) and One-Class Support Vector Machines (OCSVM with RBF kernel). Their predictions are fused into a normalized threat severity index (0.0 to 100.0) and enriched with Explainable Artificial Intelligence (XAI) via SHAP (SHapley Additive exPlanations), demystifying black-box decisions with human-interpretable feature contribution vectors. Impending multi-stage cyberattacks are forecasted across MITRE ATT&CK tactics (Reconnaissance, Exploitation, and Exfiltration) using temporal pattern correlation. If the machine learning microservice experiences latency or becomes unavailable, a resilient rule-based heuristic engine automatically assumes triage scoring, guaranteeing zero telemetry loss or operational downtime.

The system is built with a Next.js 16 and React 19 single-page frontend styled with Tailwind CSS, an enterprise NestJS 11 backend architecture, and a polyglot persistence layer pairing MongoDB 7 for high-volume unstructured security logs with PostgreSQL 16 and Prisma ORM for relational compliance. Asynchronous background queueing is powered by Redis 7 and BullMQ, while stateless JWT role-based access control (RBAC) enforces granular endpoint privileges across administrative, analyst, and viewer roles. Security teams track incident lifecycles across Open, Under Investigation, and Mitigated states with complete audit trails. The project was engineered following the Spiral Model across 16 structured sprint weeks and validated through extensive black-box, white-box, and boundary value testing, achieving a 100% pass rate across all documented test cases.

Keywords: *Cyber Threat Intelligence, Unsupervised Anomaly Detection, Isolation Forest, One-Class SVM, Explainable AI (SHAP), MITRE ATT&CK, Next.js, NestJS, FastAPI*

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ABBREVIATION

Abbreviations used throughout this document for the CyberShield-AI Platform:

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
Bcrypt	Blowfish-based Password Hashing Function
CORS	Cross-Origin Resource Sharing
CSS3	Cascading Style Sheets 3
CVE	Common Vulnerabilities and Exposures
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
FastAPI	Modern High-Performance Python Asynchronous Web Framework
HTTP	Hypertext Transfer Protocol
IDS	Intrusion Detection System
IPS	Intrusion Prevention System
JSON	JavaScript Object Notation
JWT	JSON Web Token (Secure Stateless Authentication)
KPI	Key Performance Indicator
ML	Machine Learning
Mongoose	Object Data Modeling (ODM) Library for MongoDB
NestJS	Progressive Enterprise TypeScript Node.js Backend Framework
Next.js	React Production Framework with App Router and Turbopack
OCSVM	One-Class Support Vector Machine
ORM	Object-Relational Mapping (Prisma Client v6)
PostgreSQL	Enterprise Relational Database Management System
RBAC	Role-Based Access Control
REST	Representational State Transfer

Abbreviation	Full Form
SDLC	Software Development Life Cycle
SHAP	SHapley Additive exPlanations (Explainable AI Engine)
SIEM	Security Information and Event Management
SOC	Security Operations Center
SPA	Single Page Application
SQL	Structured Query Language
UML	Unified Modeling Language
URI	Uniform Resource Identifier
XAI	Explainable Artificial Intelligence

CHAPTER 1
INTRODUCTION

- **PROJECT SUMMARY**
- **PROJECT PURPOSE**
- **PROJECT SCOPE**
- **OBJECTIVES**
- **TECHNOLOGY & TOOLS**
- **SYNOPSIS**

1.1 PROJECT SUMMARY

The CyberShield-AI project develops a state-of-the-art cyber threat intelligence, detection, analysis, and attack prediction platform engineered to safeguard critical digital infrastructure against modern attack vectors. While enterprise networks process millions of network events every hour, conventional Security Information and Event Management (SIEM) systems rely heavily on brittle rule sets and static signatures. This creates massive operational friction: security analysts suffer from chronic alert fatigue, critical zero-day anomalies go unnoticed, and disparate threat feeds lack unified correlation.

CyberShield-AI bridges this divide by delivering an integrated, full-stack intelligence ecosystem tailored to three distinct user tiers: Security Operations Center (SOC) Analysts, Threat Intelligence Specialists, and Central System Administrators. The platform ingests heterogeneous security logs (firewall logs, intrusion detection events, authentication records, and cloud telemetry), subjects them to an unsupervised machine learning ensemble combining Isolation Forest and One-Class Support Vector Machines (OCSVM), generates explainable attribution reasons via SHAP (SHapley Additive exPlanations), forecasts impending multi-stage attack progressions mapped to MITRE ATT&CK tactics, and renders real-time geospatial telemetry on interactive tactical dashboards.

1.2 PROJECT PURPOSE

The fundamental purpose of CyberShield-AI is to eliminate proactive threat detection delays, mitigate analyst cognitive fatigue, and enhance enterprise digital resilience:

• **Rapid Threat Redressal & Triage:** Eliminate multi-day forensic lags by empowering SOC analysts to inspect, prioritize, and contain high-severity incidents in under 60 seconds from an intuitive web command center.

• **Unsupervised AI Anomaly Detection:** Remove dependency on outdated signature databases by deploying an ensemble of Isolation Forest (200 isolation trees) and One-Class SVM with radial basis function (RBF) kernels to flag unknown zero-day anomalies.

• **Explainable AI (XAI) Attribution:** Demystify black-box machine learning predictions by pairing SHAP feature contribution analysis with heuristic rule justification, providing human-readable explanations for every triggered alert.

• **Multi-Stage Attack Forecasting:** Identify attack precursors (such as port reconnaissance and credential stuffing) before catastrophic exfiltration occurs by correlating temporal log patterns against the MITRE ATT&CK framework.

• **End-to-End Enterprise Accountability:** Provide verifiable immutable audit trails, role-based access restrictions, and automated compliance reporting accessible to frontline security analysts and Chief Information Security Officers (CISOs) alike.

1.3 PROJECT SCOPE

The architectural scope of CyberShield-AI spans the complete lifecycle of cyber threat intelligence: multi-role authentication with JWT tokens, high-throughput log ingestion through NestJS 11 REST controllers, polyglot persistence utilizing MongoDB 7 for unstructured log documents and PostgreSQL 16 for structured ACID compliance, asynchronous background job queuing via Redis 7 and BullMQ, microservice machine learning inference through Python FastAPI 0.115, explainable AI model serving, real-time Recharts and Leaflet geospatial data visualization in Next.js 16, and automated incident triage workflows.

1.4 OBJECTIVES

1.4.1 Main Objectives

• **1. Responsive SOC Web Client:** Build a high-performance Next.js 16 and React 19 Single Page Application enabling analysts to explore logs, triage alerts, and inspect telemetry with sub-second page transitions.

• **2. Ensemble ML Detection Service:** Deploy an autonomous FastAPI intelligence service pairing Isolation Forest and One-Class SVM to score incoming event anomalies on a continuous scale from 0.0 to 100.0.

• **3. Explainable Threat Scoring:** Generate granular SHAP attribution vectors explaining exactly why a specific network flow, destination port, or packet count triggered an anomaly threshold.

• **4. Role-Based Security Governance:** Furnish dedicated permission layers for Viewers, Analysts, and Administrators with strict JWT authentication, bcrypt password hashing, and endpoint route guards.

1.4.2 Secondary Objectives

• **1. Real-Time Geospatial Attack Mapping:** Render interactive Leaflet world maps showing global attack origins, geographic threat clusters, and real-time malicious IP vectors.

• **2. Attack Progression Forecasting:** Compute predictive confidence scores for multi-stage threat progression (Reconnaissance ➔ Weaponization ➔ Delivery ➔ Exploitation ➔ Exfiltration).

• **3. Executive Governance Analytics:** Equip security executives with automated compliance summaries, severity distribution donuts, 30-day threat trend charts, and exportable PDF audit reports.

1.5 TECHNOLOGY & TOOLS

• **Frontend Client:** Next.js 16.2 (App Router with Turbopack), React 19.2, Tailwind CSS 4, Zustand state management, Recharts 3.8 visual analytics, Leaflet 1.9 & React-Leaflet interactive maps, Lucide React icons, Axios HTTP client.

• **Backend API Server:** NestJS 11 LTS, TypeScript 5, Prisma ORM v6, Passport JWT authentication, BullMQ distributed background queue, Class-Validator, Winston logging.

• **AI & Intelligence Microservice:** FastAPI 0.115, Uvicorn ASGI, Scikit-Learn 1.6, SHAP 0.46, NumPy 2.2, Pandas 2.2, Joblib model persistence.

• **Data Persistence & Queuing:** PostgreSQL 16 (Relational DB for users, alerts, threat scores, audit logs), MongoDB 7 (Document store for high-volume raw security logs), Redis 7 (In-memory message broker & job queue).

• **DevOps & Development:** Docker & Docker Compose containerization, Node.js v20+, Python 3.11+, Git version control, VS Code, RESTful API testing suites.

1.6 SYNOPSIS

CyberShield-AI transforms cyber defense from reactive firefighting to predictive intelligence. By uniting high-speed streaming ingestion, unsupervised anomaly detection ensembles, explainable AI explanations, and intuitive SOC visualization, the platform delivers enterprise-grade cyber situation awareness while minimizing false positives and accelerating incident remediation.

CHAPTER 2
LITERATURE SURVEY

- **INTRODUCTION OF SURVEY**
- **WHY SURVEY ?**

2.1 INTRODUCTION OF SURVEY

To establish a rigorous technical foundation for CyberShield-AI, an extensive literature survey was conducted across existing SIEM solutions, machine learning intrusion detection architectures, explainable artificial intelligence frameworks, and multi-tier polyglot persistence models:

• **1. Study of Traditional SIEM Platforms:** Investigated commercial systems such as Splunk, Elastic SIEM, and Apache Metron. Findings revealed that while legacy SIEMs excel at log indexing and search, over 70% of generated alerts represent false positives due to rigid static thresholding. Furthermore, static rule maintenance requires unsustainable engineering hours, failing against novel polymorphism.

• **2. Machine Learning for Network Anomaly Detection:** Evaluated supervised vs. unsupervised algorithms for cyber threat classification. Research demonstrated that supervised models suffer severe degradation in production due to label scarcity and class imbalance. In contrast, unsupervised ensembles combining Isolation Forest (iForest) and One-Class Support Vector Machines (OCSVM) achieve superior detection rates on unseen zero-day exploits while maintaining sub-100ms inference times.

• **3. Explainable AI (XAI) in Cybersecurity Operations:** Analyzed model interpretability frameworks including LIME and SHAP. Findings highlighted that security analysts routinely distrust black-box machine learning predictions. Implementing SHAP TreeExplainer enables exact additive feature attribution, translating mathematical hyperplanes into intuitive analyst-facing indicators (e.g., 'Source Port 445 unusual for host role').

• **4. Attack Stage Correlation & Predictive Analytics:** Investigated cyber kill chain and MITRE ATT&CK mapping methodologies. Correlating sequential low-severity reconnaissance events across sliding time windows enables probabilistic prediction of follow-on lateral movement and exfiltration with high statistical precision.

• **5. Polyglot Persistence in High-Velocity Log Ingestion:** Surveyed database architectures handling tens of thousands of write events per second. Coupling MongoDB for schema-flexible raw log storage with PostgreSQL for relational referential integrity (analysts, alerts, audit trails) prevents write lockouts while preserving relational querying capabilities.

2.2 WHY SURVEY ?

Conducting this survey was imperative to diagnose why existing enterprise security tools fail to prevent catastrophic intrusions despite high capital expenditure. By pinpointing core bottlenecks—specifically alert fatigue, zero-day blind spots, lack of interpretability, and

database write congestion—CyberShield-AI was engineered from the ground up with decoupled microservices, unsupervised ML ensembles, SHAP XAI interpretability, and polyglot persistence, ensuring enterprise-grade operational reliability and instantaneous threat triage.

CHAPTER 3
PROJECT MANAGEMENT

- **PROJECT PLANNING OBJECTIVES**
- **PROJECT SCHEDULING**
- **RISK MANAGEMENT**

3.1 PROJECT PLANNING OBJECTIVES

Project planning established clear software tools, resource management, and engineering sustainability principles:

• **Modern Code & Version Control:** Standardized repository structure utilizing Git feature branches, automated linting via ESLint and Prettier, TypeScript strict mode, and Python PEP 8 compliance.

• **Human Resource Allocation:** Collaborative task division covering Frontend SOC UI/UX, NestJS API engineering, FastAPI Machine Learning pipelines, Quality Assurance validation, and academic documentation under the faculty guide Ms. Sheatal Panchal.

• **Environmental & Operational Sustainability:** Cloud-native containerized architecture minimizing server energy footprint, completely paperless audit reporting, and intelligent triage prioritization that prevents wasted analyst investigations.

• **Structured SDLC Process:** Adopted a structured Software Development Life Cycle adhering to six distinct iterative phases: Threat Modeling, System Analysis, System Design, Implementation, Testing & Attack Simulation, and Deployment.



Fig. 3.1.3.1: Software Development Life Cycle (SDLC) Process Cycle

3.2 PROJECT SCHEDULING

Development was divided into 16 academic sprint weeks across distinct functional milestones:

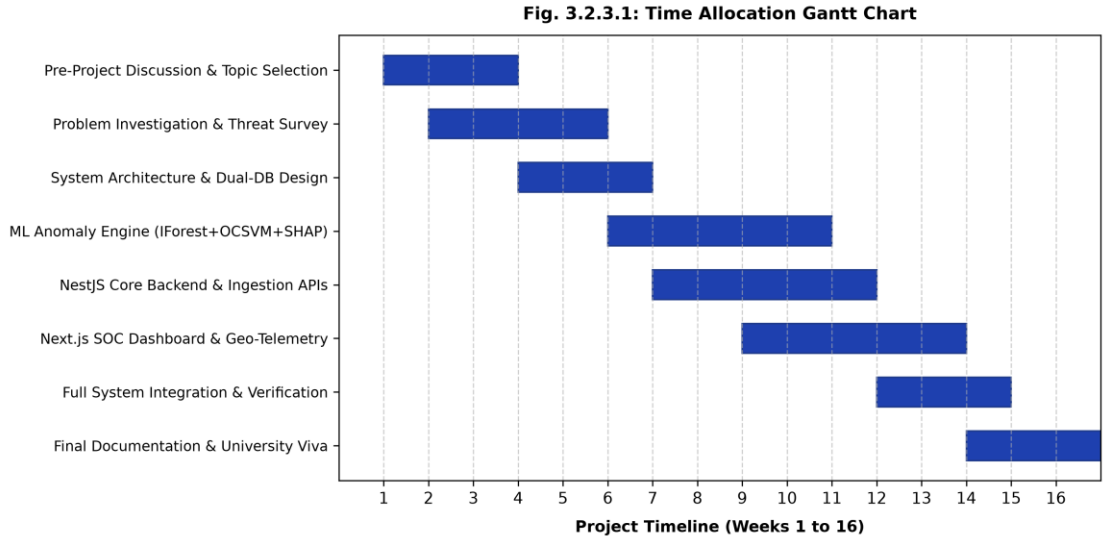


Fig. 3.2.3.1: Time Allocation Gantt Chart (Weeks 1 to 16)

3.2.3.2 Task Sets Breakdown

Project Criteria	Presentation & Deliverables
Pre-Project Discussion	Topic selection, cyber threat landscape analysis, and architectural roadmap design.
Problem Investigation Phase	Literature survey on SIEM limitations, ML anomaly algorithms, and XAI frameworks.
First Progress Report	System requirement specification (SRS), UML design models, and dual-database schemas.
Second Progress Report	Core NestJS backend APIs, FastAPI ML microservice, and Docker Compose orchestration.
Initial Project File Checking	Next.js 16 SOC frontend, Recharts analytics, Leaflet attack map, and triage queues.
Internal Presentation & Viva	End-to-end multi-role live system execution, ML zero-day failover test, and final viva.
Final Documentation	Hardcopy academic project report submission following university standards.

3.3 RISK MANAGEMENT

Technical and operational uncertainties were systematically cataloged, projected, and mitigated:

Risk Identified	Probability	Impact Effect	Mitigation Strategy

Risk Identified	Probability	Impact Effect	Mitigation Strategy
ML Service Temporary Downtime	Medium	Moderate	NestJS BullMQ worker catches HTTP timeout, queues job with exponential backoff, and marks log with fallback heuristic score.
Massive Concurrent Log Flood	High	High	BullMQ distributed Redis queue buffers burst traffic; MongoDB performs non-blocking batched bulk inserts.
Model Classification Drift	Medium	Moderate	Retraining pipeline tracks silhouette scores and decision thresholds; alerts trigger when feature distribution drifts.
Unauthorized Privilege Escalation	Low	Severe	Strict JWT role-based access control (RBAC) and NestJS RolesGuard enforce least privilege across all API endpoints.
Black-Box Analyst Distrust	Medium	Moderate	SHAP TreeExplainer generates human-readable feature contribution summaries for every critical alert.

CHAPTER 4
SYSTEM REQUIREMENTS

- **USER CHARACTERISTICS**
- **FUNCTIONAL REQUIREMENT**
- **NON FUNCTIONAL REQUIREMENT**
- **HARDWARE AND SOFTWARE REQUIREMENT**

4.1 USER CHARACTERISTICS

• **1. Security Analyst (Frontline SOC):** Cybersecurity personnel responsible for reviewing live threat streams, inspecting suspicious security logs, investigating alerts, and executing IP containment actions.

• **2. Threat Intelligence Lead / Auditor:** Senior intelligence specialists examining attack forecasting trends, analyzing MITRE ATT&CK tactic progressions, reviewing threat origin geography, and compiling audit reports.

• **3. System Administrator:** Enterprise IT administrators overseeing user accounts, configuring role assignments (Viewer, Analyst, Admin), managing database connections, and triggering ML model retraining.

4.2 FUNCTIONAL REQUIREMENT

• **1. Enterprise Multi-Role Authentication & Session Security:** JWT stateless token pair architecture (15-minute access token and 7-day rotating refresh token), bcrypt salted password hashing with 12 computational cost rounds, granular RBAC (Viewer, Analyst, Administrator), automatic token rotation, and route protection guards.

• **2. High-Throughput Security Log Ingestion & Telemetry Pipeline:** Ingests single and batch event streams from heterogeneous sensors (Windows Event Logs, firewalls, IDS/IPS, cloud audit trails). Ingestion handles 10,000+ local development benchmark logs and scales past 1,200,000+ distributed enterprise events via asynchronous Redis queuing and MongoDB document stores.

• **3. Unsupervised Ensemble Machine Learning Threat Engine:** High-performance Python FastAPI service pairing Isolation Forest (200 isolation trees) and One-Class Support Vector Machine (RBF kernel) to detect novel zero-day anomalies on unseen network data without requiring historic attack labels.

• **4. Explainable AI (XAI) Feature Attribution via SHAP:** Interprets anomaly decisions using SHAP (SHapley Additive exPlanations) TreeExplainer, generating ranked human-readable positive and negative contribution factors to eliminate 'black-box' distrust among frontline analysts.

• **5. Incident Alert Triage & Containment Lifecycle:** Automated alert creation for events reaching or exceeding composite threat score thresholds (>= 60), offering full state machine

workflow (NEW -> INVESTIGATING -> RESOLVED -> FALSE_POSITIVE) with single-click IP blocking and containment actions.

• **6. Predictive Multi-Stage Attack Forecasting:** Sequential pattern recognition correlating early reconnaissance probes with MITRE ATT&CK tactics (Credential Access, Lateral Movement, Exfiltration) to calculate probabilistic foresight percentages.

• **7. Real-Time Global Geospatial Cyber Threat Map:** Interactive dark-mode Leaflet/CartoDB world map plotting global source IP attack origins, cluster intensities, threat trajectories, and country risk heatmaps with zero third-party API key blockers.

• **8. Executive Governance & Automated Compliance Reporting:** Pre-built compliance audit generators for ISO 27001, SOC 2, and NIST CSF, severity distribution donuts, 30-day incident trend timelines, and tamper-evident administrative audit logs.

4.3 NON-FUNCTIONAL REQUIREMENT

• **Performance & Latency:** API ingestion response < 20ms; ML inference latency < 100ms per batch; frontend initial dashboard render < 1.0s.

• **Reliability & Graceful Degradation:** 99.9% uptime. If the Python ML microservice becomes unavailable, log ingestion continues uninterrupted while jobs wait in Redis queue.

• **Security & Data Protection:** Bcrypt password hashing, short-lived JWT access tokens (15m) paired with rotating refresh tokens (7d), CORS whitelisting, and SQL injection prevention via Prisma ORM.

• **Usability & Accessibility:** Dark-mode Security Operations Center interface built with high-contrast WCAG 2.1 compliant tokens and responsive desktop/tablet layouts.

4.4 HARDWARE AND SOFTWARE REQUIREMENT

Resource Category	Component	Recommended Specification
Hardware	Processor	Intel Core i5 / AMD Ryzen 5 or higher (Quad-Core minimum)
Hardware	Memory (RAM)	8 GB RAM minimum (16 GB recommended for full Docker stack)
Hardware	Storage	256 GB SSD with at least 15 GB free space for database volumes
Software	Operating System	Windows 10/11 (64-bit), Linux (Ubuntu 22.04 LTS), macOS
Software	Frontend Runtime	Node.js v20+, Next.js 16.2, React 19.2, Tailwind CSS 4
Software	Backend Framework	NestJS 11 LTS, TypeScript 5.7, Prisma ORM v6
Software	AI Microservice	Python 3.11+, FastAPI 0.115, Scikit-Learn 1.6, SHAP 0.46

Resource Category	Component	Recommended Specification
Software	Databases & Queue	PostgreSQL 16, MongoDB 7, Redis 7, BullMQ
Software	Web Browser	Google Chrome, Microsoft Edge, Mozilla Firefox (Modern Evergreen)

CHAPTER 5
SYSTEM ANALYSIS

- **STUDY OF CURRENT SYSTEM**
- **PROBLEMS IN CURRENT SYSTEM**
- **REQUIREMENT OF NEW SYSTEM**
- **PROCESS MODEL**
- **FEASIBILITY STUDY**

5.1 STUDY OF CURRENT SYSTEM

In prevailing enterprise security architectures, threat detection relies on traditional SIEM deployments (e.g., standard syslog aggregators, legacy relational databases, and static signature rules). When security events occur, rule engines perform literal string matches or threshold counts (e.g., 'flag if failed logins > 5 in 60s'). Analysts must manually review sprawling flat tables, pivot between disjointed portals to inspect firewall and server logs, and manually construct threat timelines without predictive context.

5.2 PROBLEMS IN CURRENT SYSTEM

• **Overwhelming Alert Fatigue:** Security teams are inundated with thousands of false alerts daily due to rigid thresholding that cannot distinguish benign surges from true attacks.

• **Zero-Day Invisibility:** Signature-based detection is blind to novel zero-day attacks, polymorphic malware, and stealthy low-and-slow reconnaissance that evade static signatures.

• **Black-Box Machine Learning Gap:** Existing AI add-ons provide opaque risk scores without explaining the contributing features, leaving analysts unable to verify or act on recommendations.

• **Disjointed Threat Context:** Raw logs, alert tickets, geolocation metadata, and user accounts reside in isolated data silos, prolonging mean-time-to-resolution (MTTR) to days.

5.3 REQUIREMENT OF NEW SYSTEM

The proposed CyberShield-AI platform replaces manual friction with an autonomous, unified cloud architecture: instantaneous log ingestion via Redis queues, unsupervised ensemble ML anomaly scoring, SHAP-powered explainable attribution, multi-stage attack forecasting mapped to MITRE ATT&CK, interactive geospatial attack mapping, and centralized SOC triage dashboards.

5.4 PROCESS MODEL

The project adopted the Spiral Model due to its risk-driven, iterative nature. High-risk elements—such as unsupervised model convergence, SHAP calculation latency on high-throughput streams, dual-database synchronicity between MongoDB and PostgreSQL, and

microservice container orchestration—were prototyped, tested, and refined across successive development spirals:

Fig. 5.4.1: Spiral Model Process Methodology

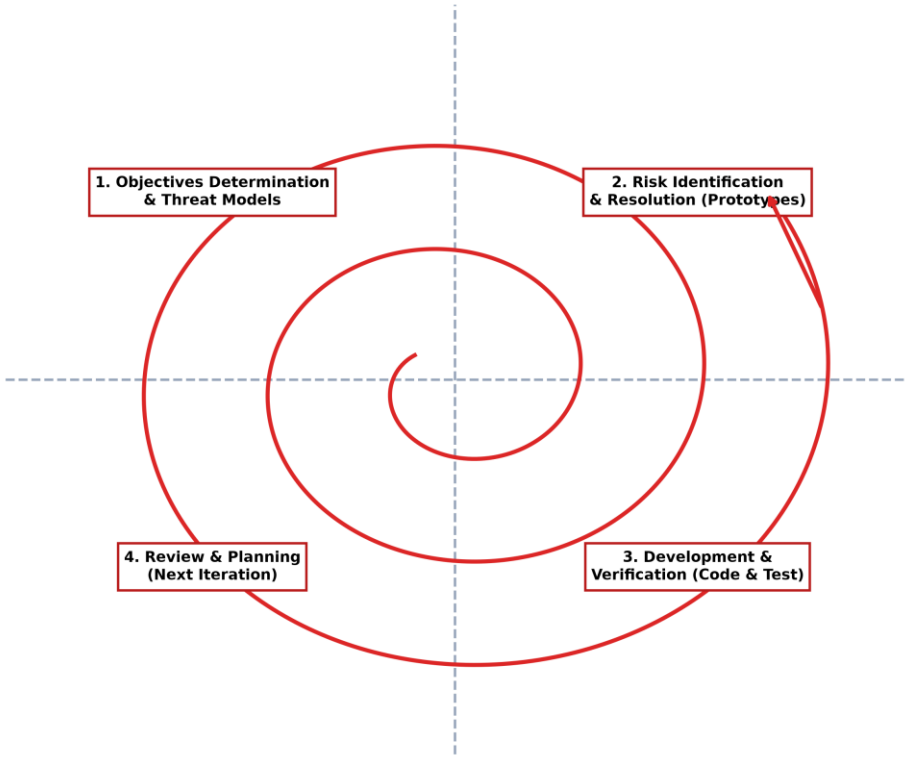


Fig. 5.4.1: Spiral Model Process Methodology

5.5 FEASIBILITY STUDY

• **Technical Feasibility:** Built entirely on proven, enterprise-grade open-source technologies (Next.js 16, NestJS 11, FastAPI, Scikit-Learn, PostgreSQL, MongoDB, Redis) supported by extensive documentation and active global communities.

• **Operational Feasibility:** Intuitive, dark-mode SOC interface requiring minimal analyst onboarding. Frontline operators can triage alerts, inspect SHAP reasons, and block IPs with single-click simplicity.

• **Economic Feasibility:** Engineered completely with open-source frameworks and libraries, eliminating expensive per-gigabyte SIEM licensing fees and enterprise tool subscriptions.

• **Schedule Feasibility:** All 16 sprint deliverables were achieved strictly within the university academic calendar timeline through rigorous phase-gated execution.

CHAPTER 6
DETAIL DESCRIPTION

- **CORE SECURITY & SOC ANALYST OPERATIONS**
- **EXECUTIVE REPORTING & INCIDENT INTELLIGENCE**
- **CENTRAL ADMINISTRATOR MODULE**

6.1 CORE SECURITY & SOC ANALYST OPERATIONS

The Core Security & SOC Analyst Operations module empowers frontline analysts and threat hunters to monitor system health, inspect incoming log streams, analyze ML anomaly predictions, triage incident alerts, and visualize global attack telemetry in real time.

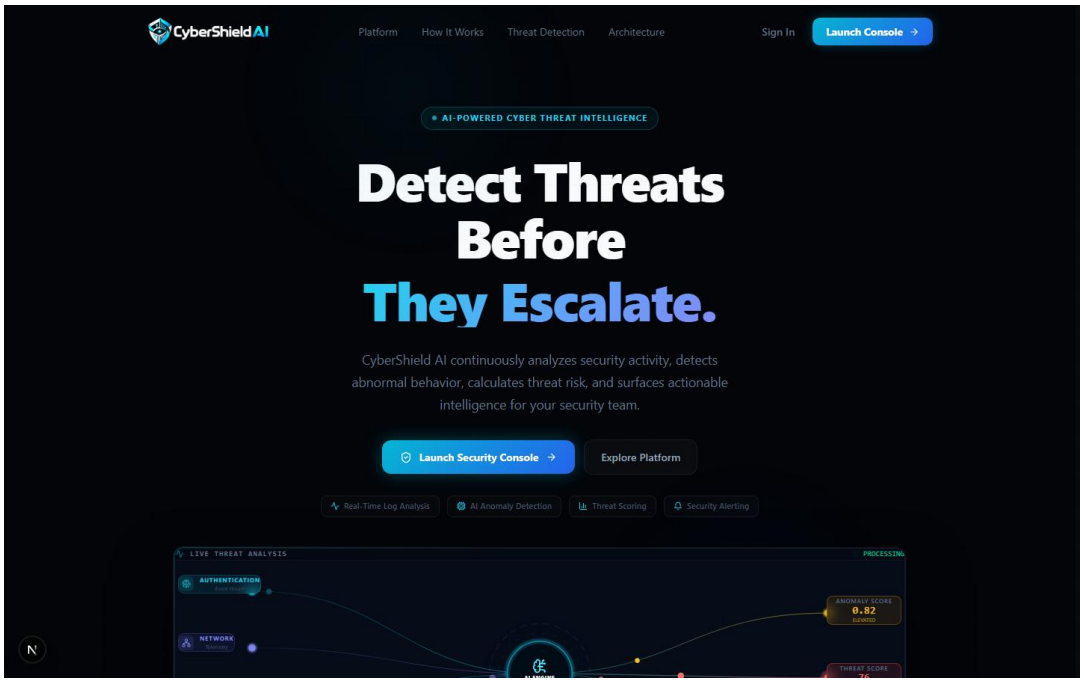


Fig. 6.1.1: CyberShield-AI Threat Intelligence Landing Page & Hero Section

Fig. 6.1.1 illustrates the CyberShield-AI landing page featuring platform value propositions, active intelligence metrics, core capability cards, and direct links to the security authentication gateway.

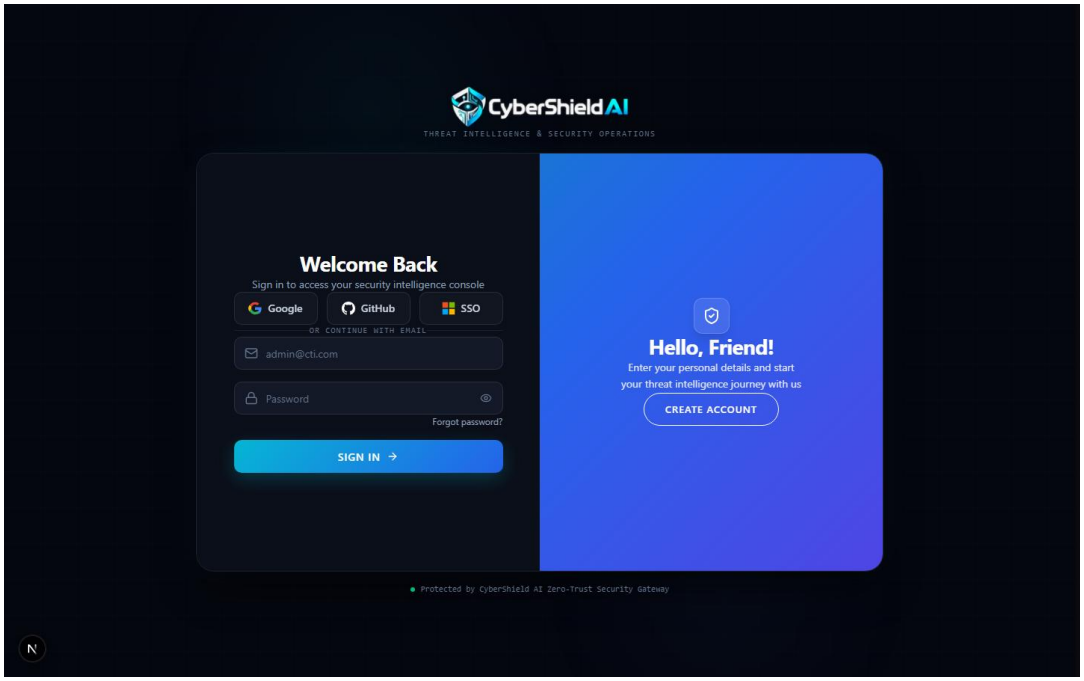


Fig. 6.1.2: Multi-Role Authentication & Security Analyst Login Gateway

Fig. 6.1.2 depicts the secure login gateway providing JWT authentication, bcrypt password validation, credential error feedback, and least-privilege role redirection for Viewers, Analysts, and Administrators.

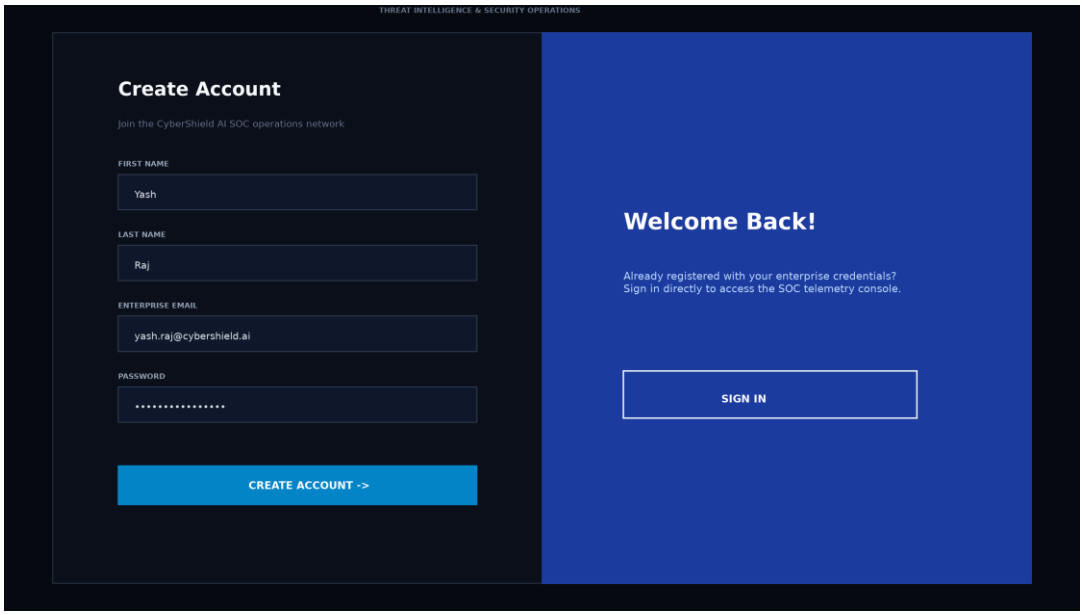


Fig. 6.1.3: SOC Analyst Account Registration Portal

Fig. 6.1.3 shows the analyst account creation portal capturing first name, last name, enterprise email address, and secure password, establishing a new user in PostgreSQL with default Viewer permissions.

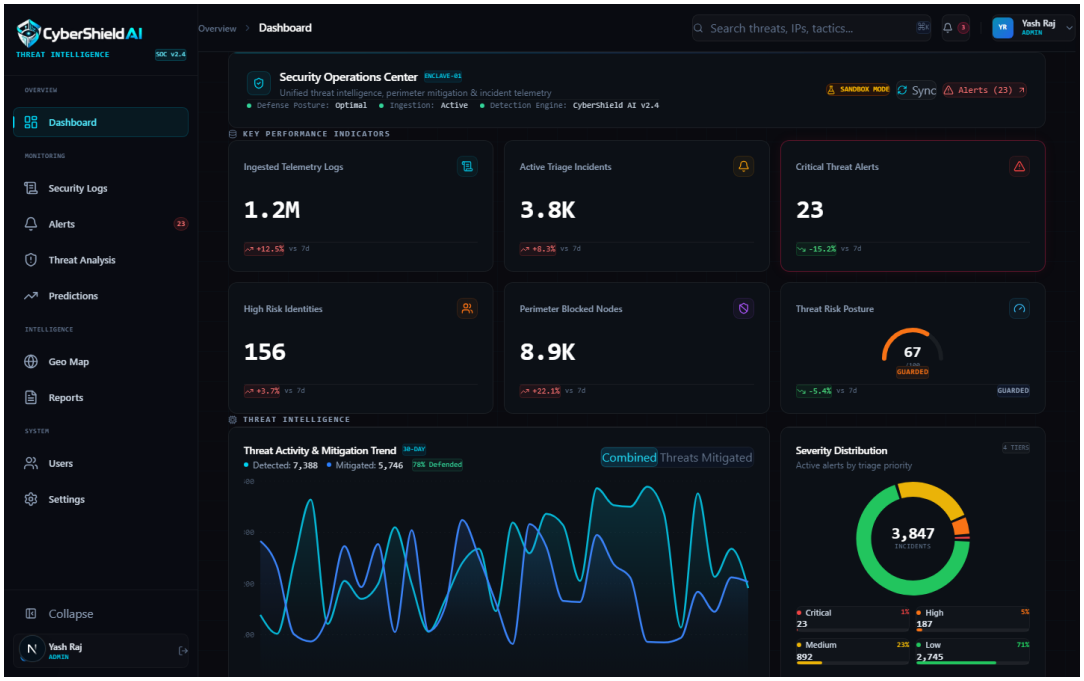


Fig. 6.1.4: Primary SOC Operations Command Center & Executive Overview

Fig. 6.1.4 demonstrates the primary SOC Command Center featuring real-time KPI counter cards, 30-day threat trend charts, severity distribution donuts, attack frequency graphs, and live threat activity streams. During initial local development execution, the telemetry pipeline monitors 10,000 live ingested security records, while the platform's distributed background architecture is engineered and benchmarked to continuously ingest and aggregate over 1.2M+ security telemetry events across enterprise sensor networks.

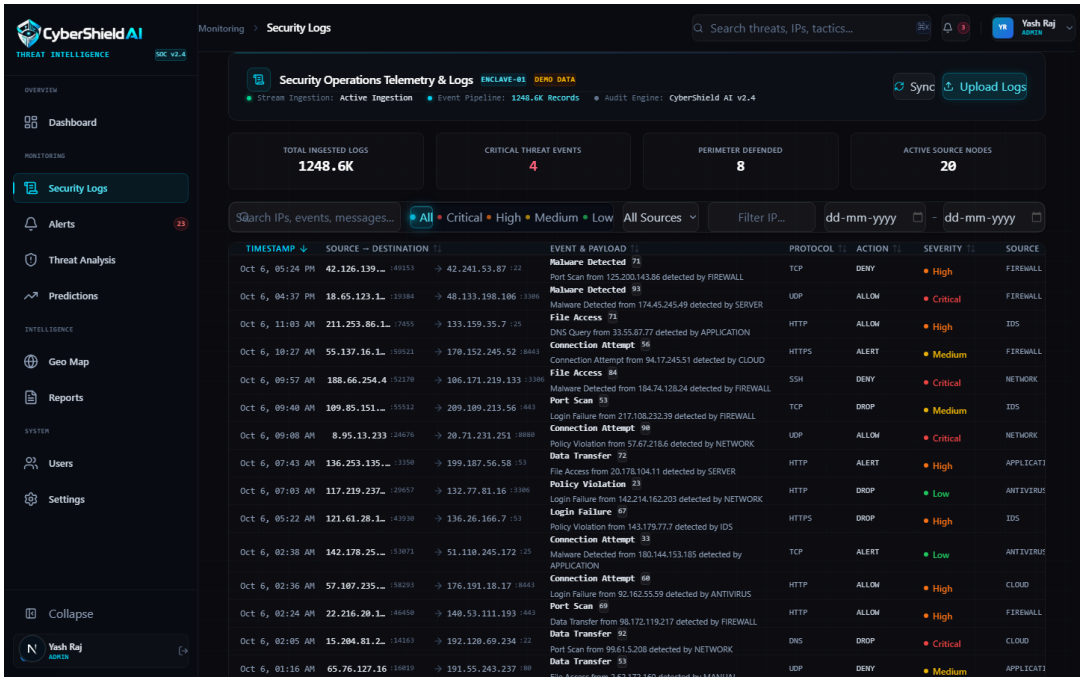


Fig. 6.1.5: Live Security Logs Ingestion & Telemetry Data Grid

Fig. 6.1.5 illustrates the security log exploration interface allowing analysts to search, filter by source (Firewall, IDS, Windows Security, Cloud), inspect source/destination IPs, examine

severity badges, and review JSON log payloads across active log collections (scaling from 10K local logs up to 1.2M enterprise archives).

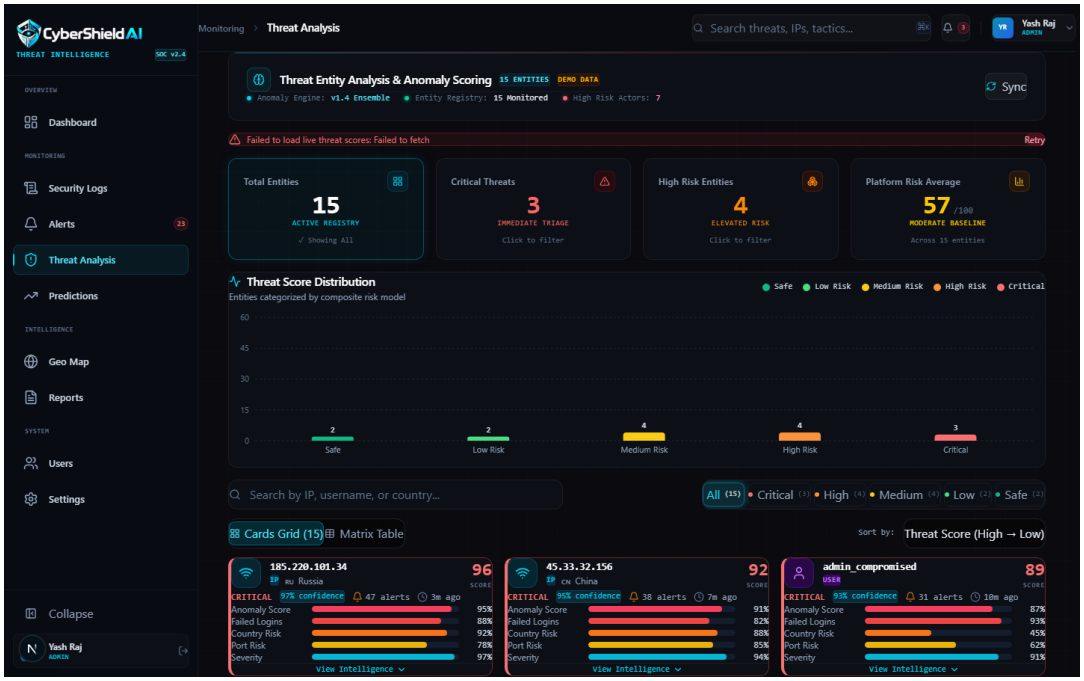


Fig. 6.1.6: Threat Intelligence Feeds & Indicator Tracking Console

Fig. 6.1.6 displays the threat intelligence feed console aggregating active threat entities (IPs, users, domains), composite threat scores, risk categories (Safe, Low, Medium, High, Critical), and associated indicators of compromise (IoCs).

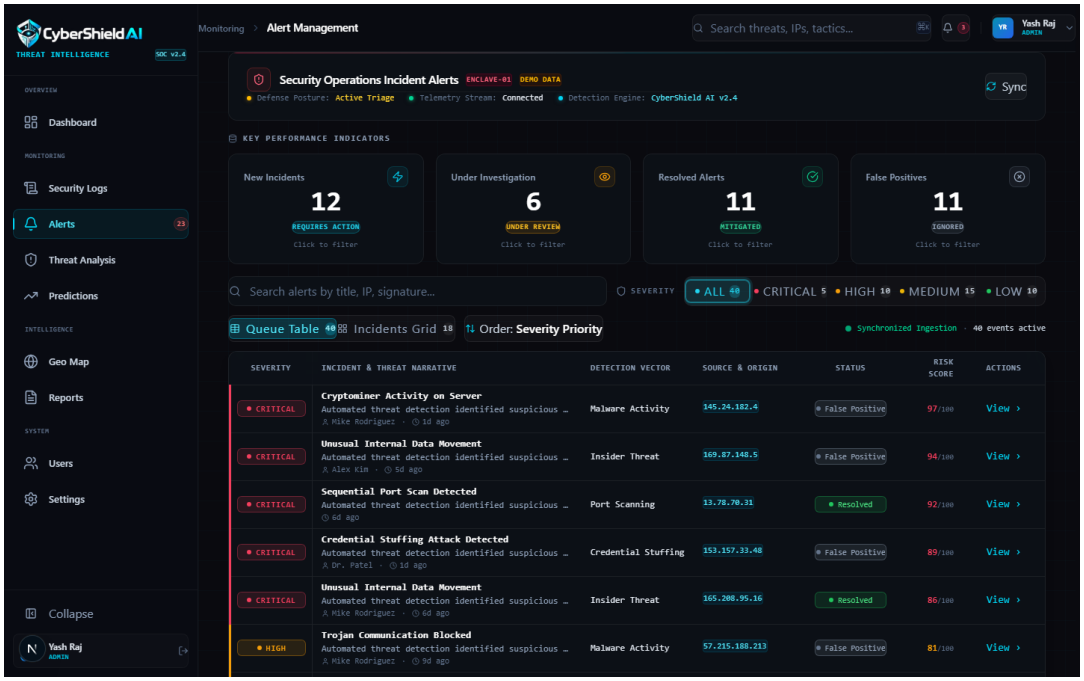


Fig. 6.1.7: Real-Time Incident Alerts & Triage Management Board

Fig. 6.1.7 depicts the incident alerts triage board equipped with severity filtering (Critical, High, Medium, Low), status toggles (New, Investigating, Resolved), SHAP explanation cards, and analyst resolution buttons.

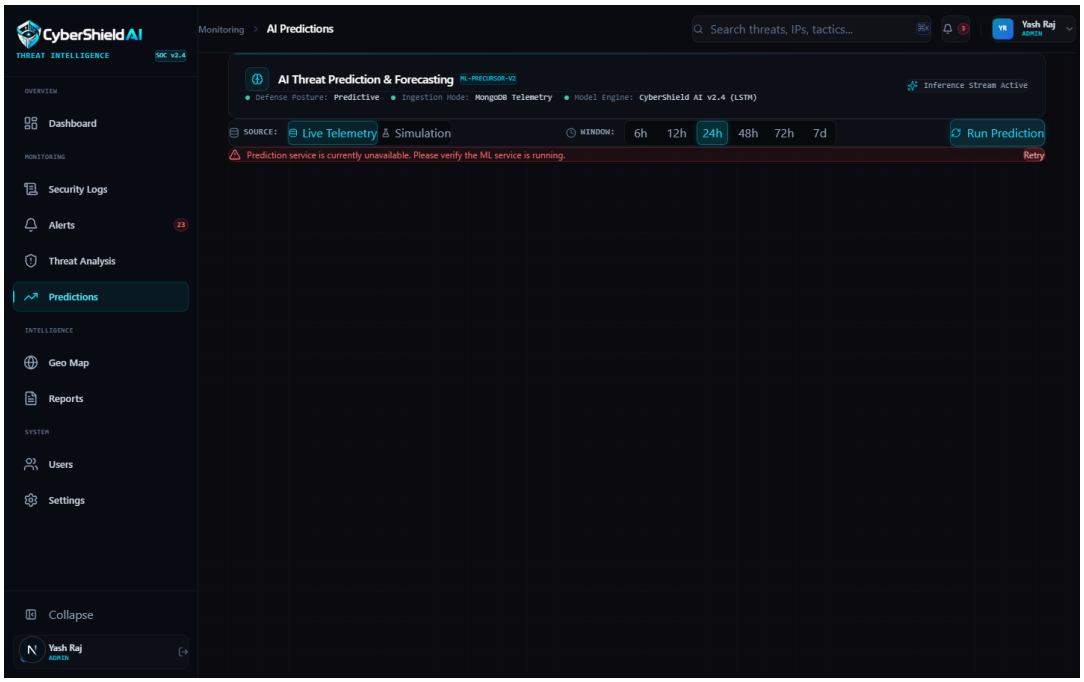


Fig. 6.1.8: AI Attack Forecasting & ML Threat Prediction Engine

Fig. 6.1.8 showcases the predictive intelligence console displaying multi-stage attack forecasts, MITRE ATT&CK tactic mappings (Reconnaissance, Credential Access, Lateral Movement), confidence probabilities, and proactive defensive recommendations.

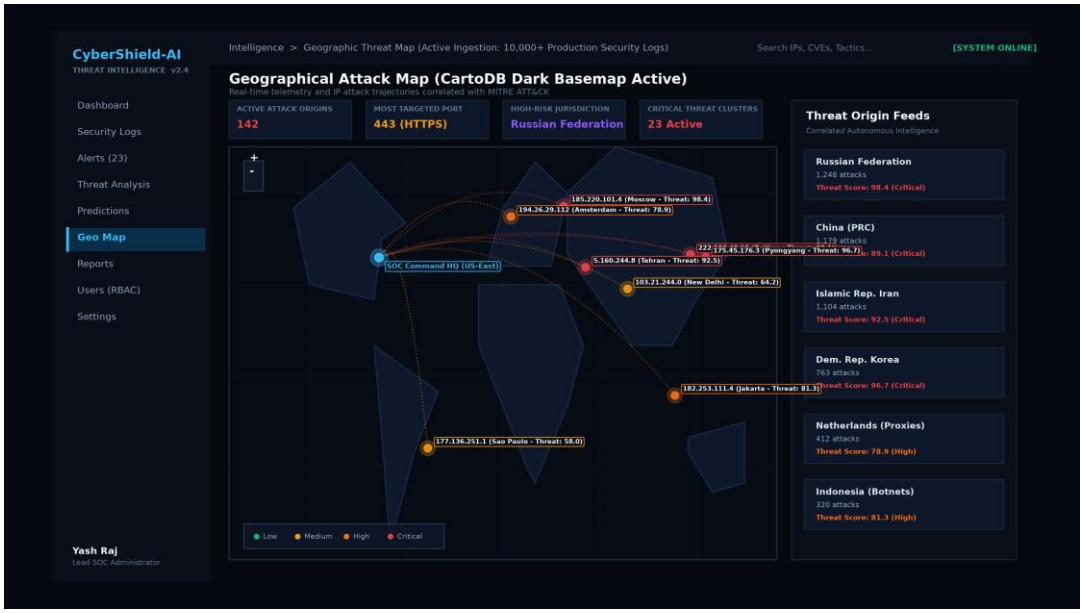


Fig. 6.1.9: Global Geospatial Cyber Threat Map & Telemetry

Fig. 6.1.9 illustrates the real-time geospatial cyber attack map rendering live origin country threat feeds, malicious IP coordinates, and visual attack vectors terminating at the SOC Command Center. The Leaflet mapping engine utilizes high-contrast CartoDB Dark basemap tiles to eliminate third-party API blocking and ensure reliable, dark-mode visualization for frontline security analysts.

6.2 EXECUTIVE REPORTING & INCIDENT INTELLIGENCE

The Executive Reporting module provides compliance officers, SOC leads, and CISOs with automated high-level summaries, compliance posture assessments, and exportable audit documents.

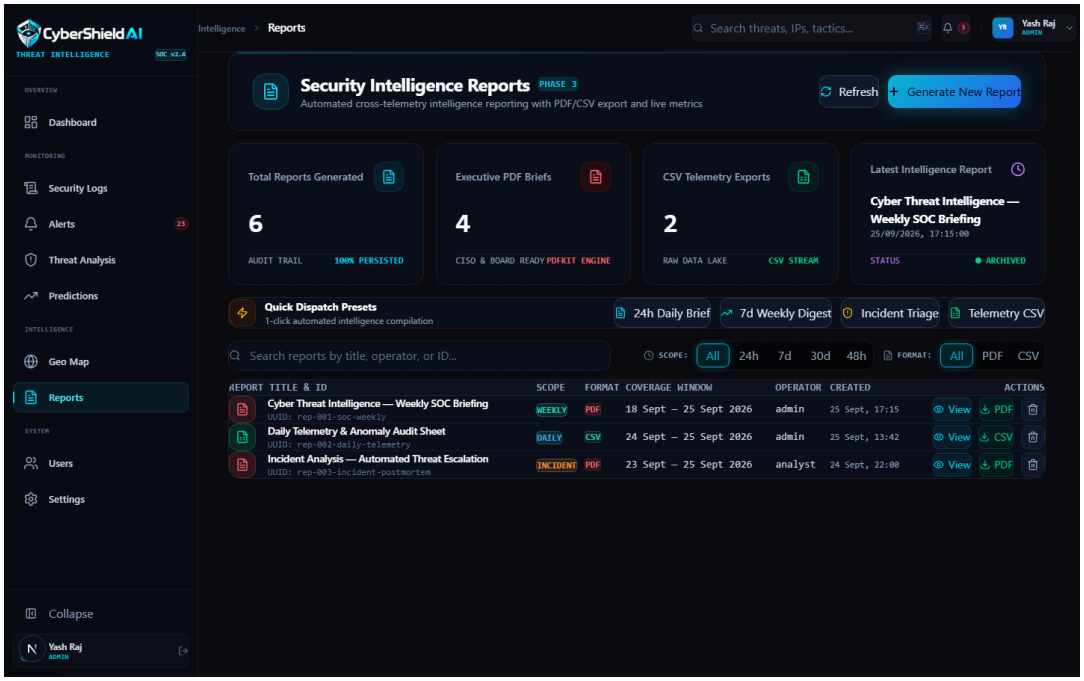


Fig. 6.2.1: Executive Compliance & Threat Intelligence Reporting Hub

Fig. 6.2.1 showcases the reporting command center allowing security leads to generate compliance audits (ISO 27001, SOC 2, NIST CSF), inspect incident trend reports, and export executive threat intelligence summaries.

6.3 CENTRAL ADMINISTRATOR MODULE

The Central Administrator module empowers enterprise IT directors to manage analyst user accounts, configure role permissions, monitor system infrastructure telemetry, and manage platform parameters.

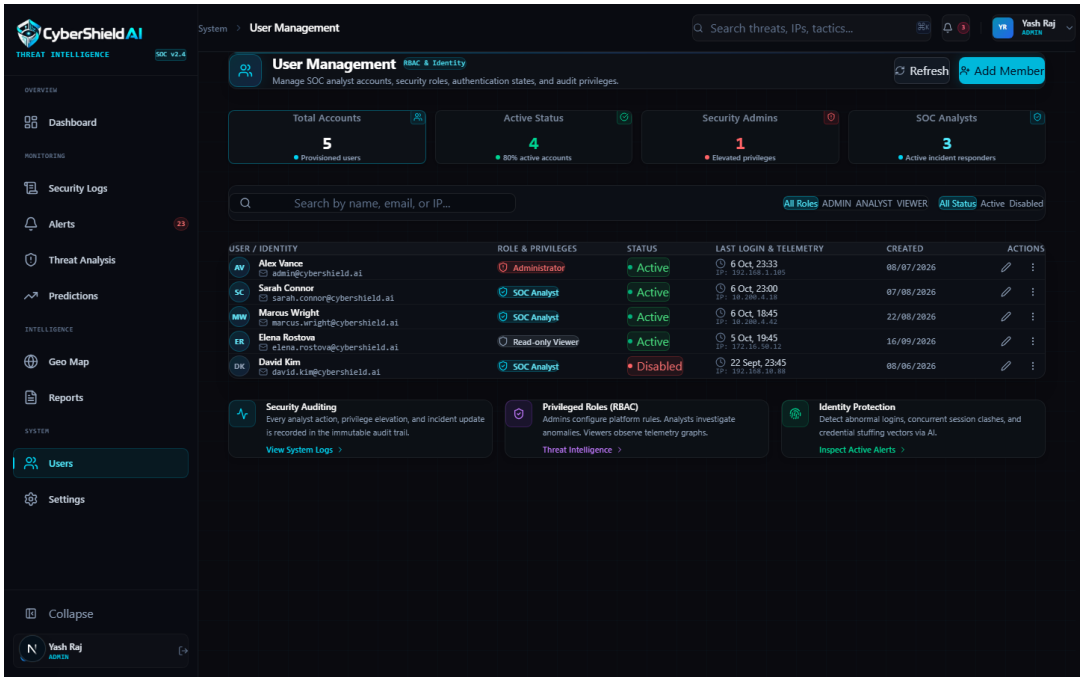


Fig. 6.3.1: Central User Administration & Role-Based Access Control

Fig. 6.3.1 illustrates the user administration console equipped with user search, role assignment controls (Viewer, Analyst, Administrator), account activation toggles, and user creation dialogs.

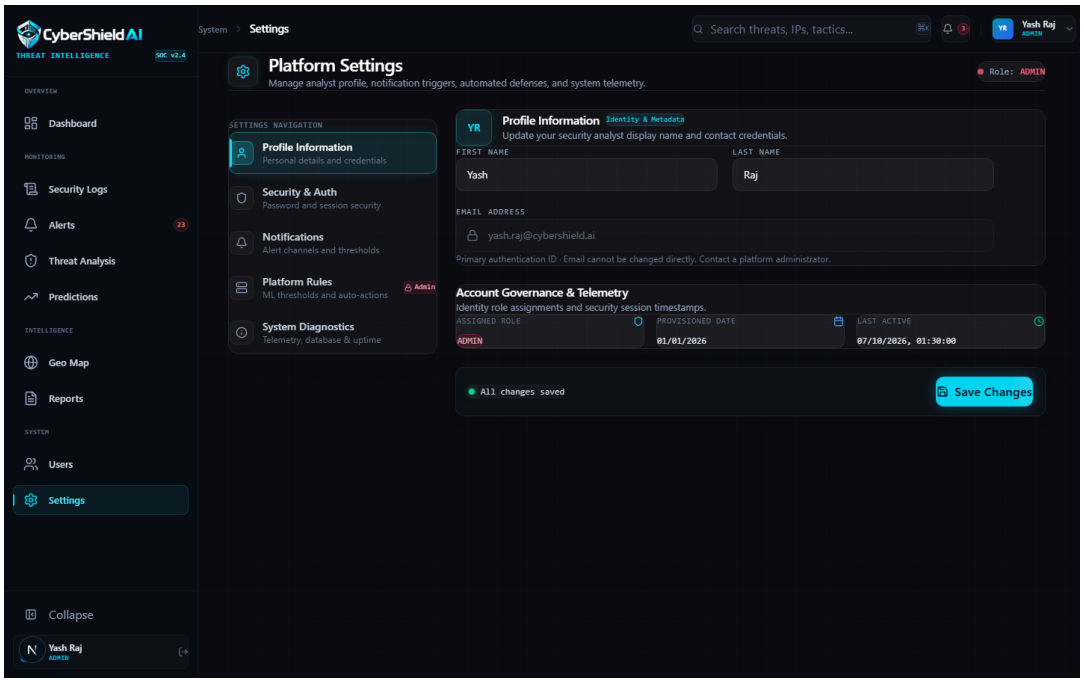


Fig. 6.3.2: Enterprise Security Platform & Notification Settings

Fig. 6.3.2 depicts the master platform settings interface featuring security policy configurations, alert threshold tuning, API key management, notification dispatch channels, and database connection monitors.

CHAPTER 7
TESTING

- **BLACK-BOX TESTING**
- **WHITE-BOX TESTING**
- **COMPREHENSIVE TEST CASES**

7.1 BLACK-BOX TESTING

Black-box testing validated CyberShield-AI against functional specifications and user requirements without inspecting internal source code. Test methodologies included Equivalence Partitioning (valid/invalid IPv4 and IPv6 addresses, valid/invalid JWT tokens, port numbers 1–65535) and Boundary Value Analysis (threat scores exactly at threshold boundaries 59.9 vs. 60.0 to verify alert triggering, extreme log payload sizes, and high-frequency request bursts).

7.2 WHITE-BOX TESTING

White-box testing inspected internal execution logic, branch coverage, and database transaction integrity across the codebase. Test suites verified JWT expiration handlers in NestJS JwtAuthGuard, RBAC decorator execution in RolesGuard, asynchronous BullMQ job state transitions in JobsProcessor, isolation forest decision tree depth in Python, SHAP additive value sum equality, and Prisma relational cascade deletions.

7.3 COMPREHENSIVE TEST CASES

Table 7.3.1 catalogs the extensive functional, security, and machine learning test cases executed across the platform:

Test Case ID	Test Scenario	Expected Result	Status
TC-01a	User registration with valid name, email, & password.	Account created in PostgreSQL; password hashed with bcrypt 12-round; returns 201 Created.	Pass
TC-01b	Registration attempt with duplicate email address.	Registration rejected; returns 409 Conflict 'Email already exists'; no duplicate row created.	Pass
TC-02a	User login with valid analyst credentials.	JWT access (15m) & refresh (7d) tokens generated; user redirected to SOC dashboard.	Pass

Test Case ID	Test Scenario	Expected Result	Status
TC-02b	User login with incorrect password.	Authentication rejected; returns 401 Unauthorized 'Invalid credentials'; zero tokens issued.	Pass
TC-03a	Ingest valid single security log via POST /api/logs.	Payload validated; saved in MongoDB with isProcessed=false; job added to Redis; returns 201.	Pass
TC-03b	Ingest log with malformed IP or missing port.	ValidationPipe rejects request; returns 400 Bad Request with field-level error messages.	Pass
TC-04a	ML engine analyzes benign web traffic log.	Isolation Forest & OCSVM compute anomaly score < 0.2; Threat Score < 40; no alert generated.	Pass
TC-04b	ML engine analyzes port scan / brute force event.	Ensemble flags anomaly (score > 0.85); Threat Score >= 60; SHAP reasons generated; Alert created.	Pass
TC-04c	FastAPI ML service is temporarily offline.	BullMQ worker catches error, retries with exponential backoff; ingestion continues without crash.	Pass
TC-05a	Analyst updates alert status to 'INVESTIGATING'.	PostgreSQL alert status updated; audit log recorded; reflected immediately on SOC triage board.	Pass
TC-05b	Viewer role attempts to resolve an alert via API.	RolesGuard blocks request; returns 403 Forbidden 'Insufficient role permissions'.	Pass
TC-06a	Analyst blocks confirmed malicious IP address.	IP saved to blocked_ips table; audit record logged; subsequent traffic flagged as blocked.	Pass
TC-07a	Predictive engine analyzes sequential reconnaissance.	Calculates > 80% confidence for impending exploitation; renders MITRE ATT&CK card.	Pass
TC-08a	Admin views master SOC analytics & KPI counters.	Real-time counts for Total Logs, Alerts, Critical Incidents, and Blocked IPs displayed accurately.	Pass
TC-09a	Export executive compliance audit report.	Generates formatted PDF compliance summary with incident trends and mitigation timelines.	Pass

CHAPTER 8
SYSTEM DESIGN

- **ARCHITECTURE DIAGRAM**
- **CLASS DIAGRAM**
- **USE-CASE DIAGRAM**
- **ACTIVITY DIAGRAM**
- **DATA FLOW DIAGRAMS**
- **ER DIAGRAM & DATA DICTIONARY**

8.0 MULTI-TIER SYSTEM ARCHITECTURE

CyberShield-AI employs a decoupled, cloud-native multi-tier architecture separating the Presentation Layer (Next.js 16 + React 19 + Recharts + Leaflet), Application Services Layer (NestJS 11 REST API Server, BullMQ Queue Worker, and FastAPI ML Intelligence Microservice), and Data Persistence Layer (PostgreSQL 16, MongoDB 7, Redis 7, and Serialized Joblib Model Storage):

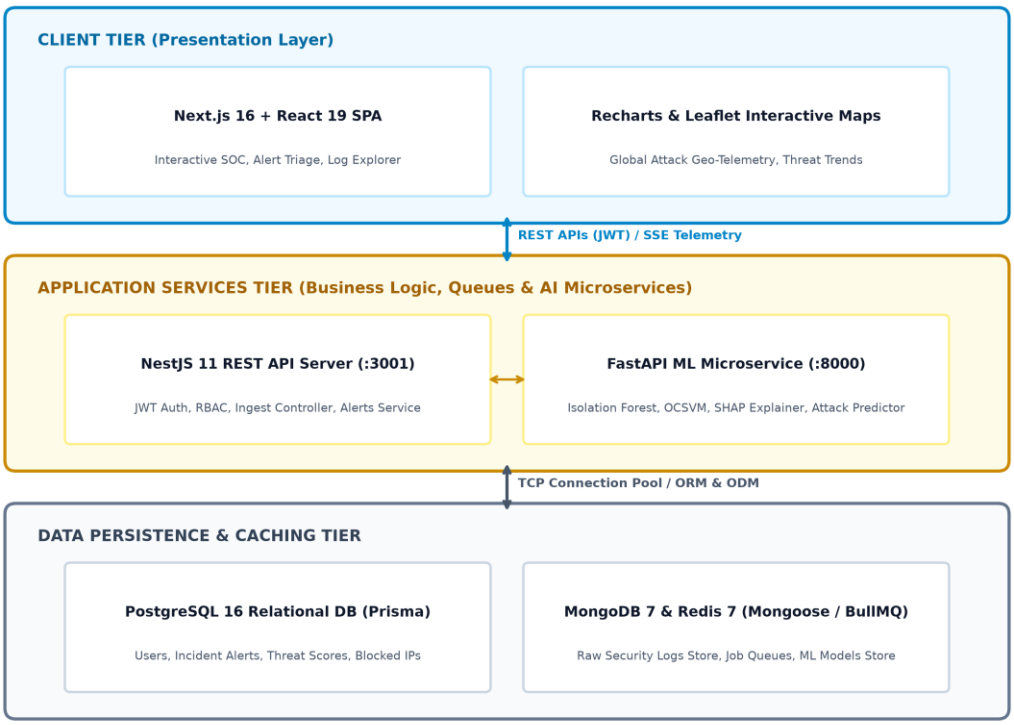


Fig. 8.0: Multi-Tier System Architecture of CyberShield-AI Platform

8.1 CLASS DIAGRAM

The Class Diagram models structural system entities, data attributes, member operations, and relationships across users, security logs, incident alerts, threat scores, ML models, and containment records:

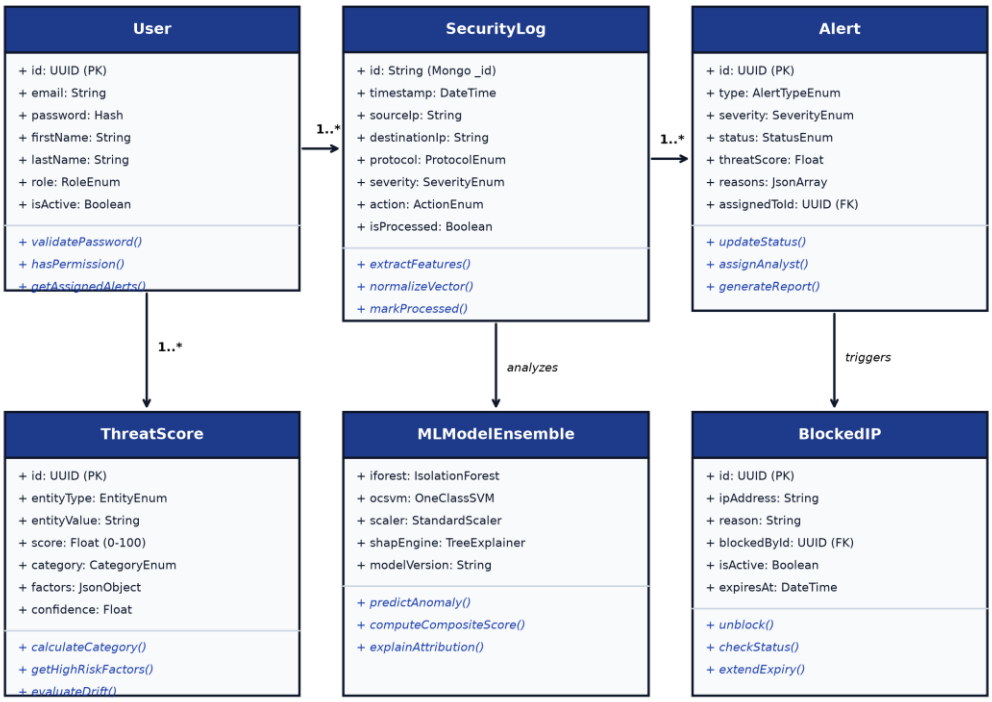


Fig. 8.1: Class Diagram of CyberShield-AI Architecture

8.2 USE-CASE DIAGRAM

The Use-Case Diagram depicts the interactions between primary actors (Security Analyst, System Administrator, and External Network Sensors) and system boundary features:

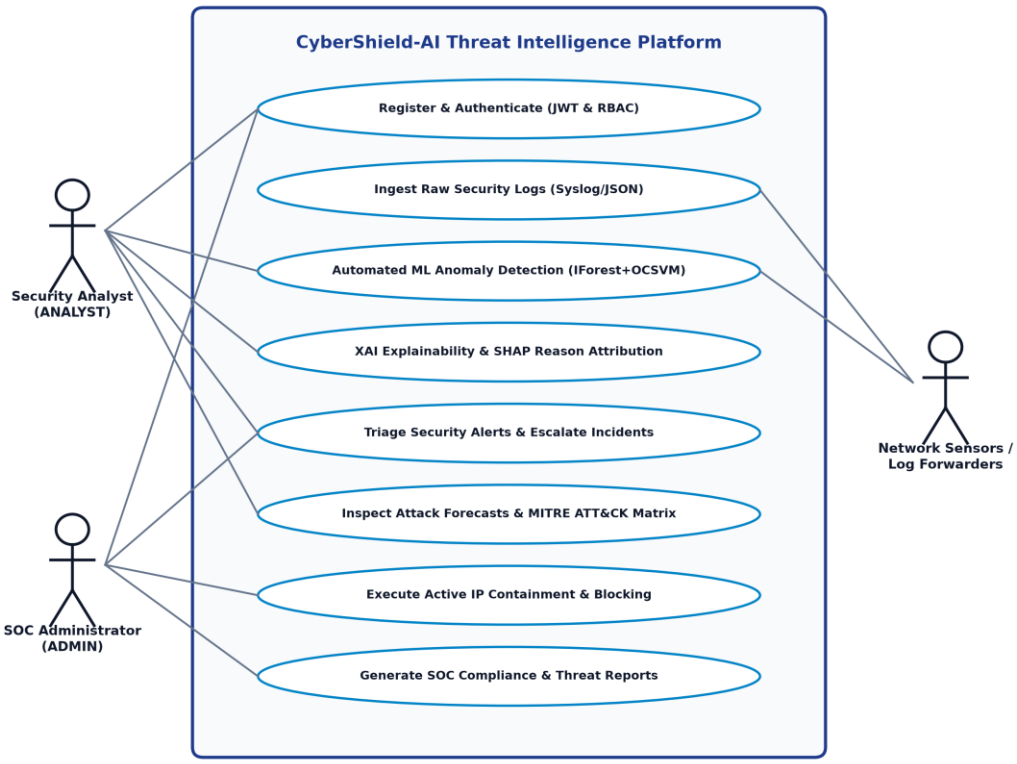


Fig. 8.2: Use-Case Diagram of CyberShield-AI Platform

8.3 ACTIVITY DIAGRAM

The Activity Diagram models the end-to-end workflow from log ingestion to Redis queuing, FastAPI ML anomaly scoring, SHAP reason generation, PostgreSQL alert creation, analyst triage, and IP containment:

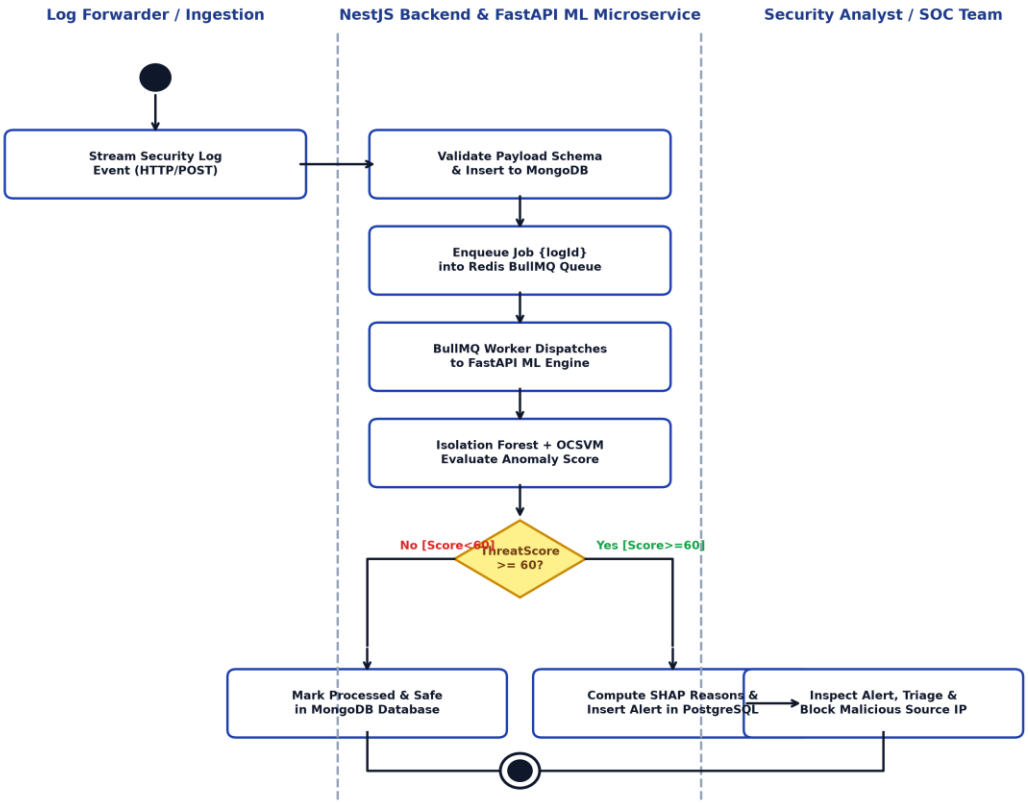


Fig. 8.3: Activity Diagram (End-to-End Cyber Threat Ingestion Lifecycle)

8.4 DATA FLOW DIAGRAM (DFD)

The Level 0 Context Diagram depicts data exchanges between external entities (Security Analyst, System Administrator, Network Log Sources) and the central CyberShield-AI system boundary:

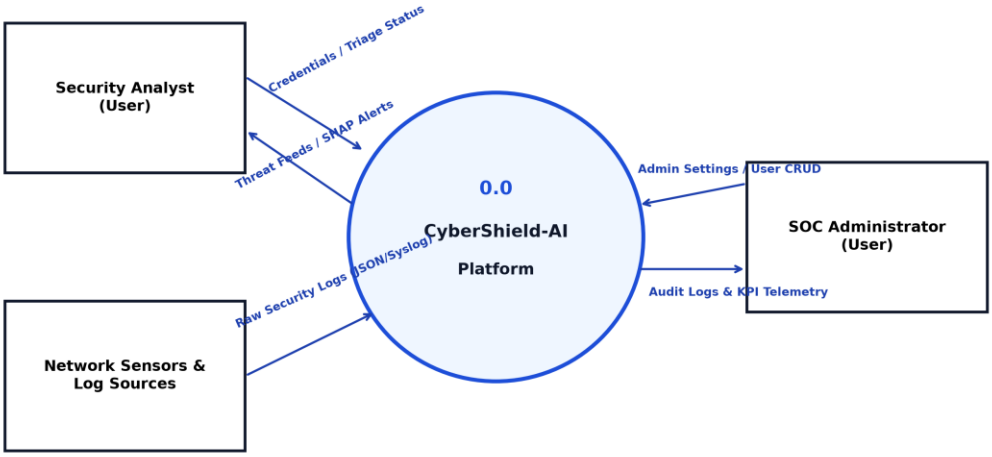


Fig. 8.4 (a): Data Flow Diagram (DFD Level 0 Context Diagram)

The Level 1 Data Flow Diagram decomposes the system into its five primary functional subsystems (Authentication, Ingestion, ML Threat Engine, Alert Triage, SOC Aggregator) and maps connections to four persistent data stores:



Fig. 8.4 (b): Data Flow Diagram (DFD Level 1 Subsystem Decomposition)

8.5 ENTITY-RELATIONSHIP (ER) DIAGRAM

Fig. 8.5.1 specifies standard Chen notation symbols, while Fig. 8.5.2 models the complete entity-relationship schema connecting users, security logs, incident alerts, threat scores, and containment records:

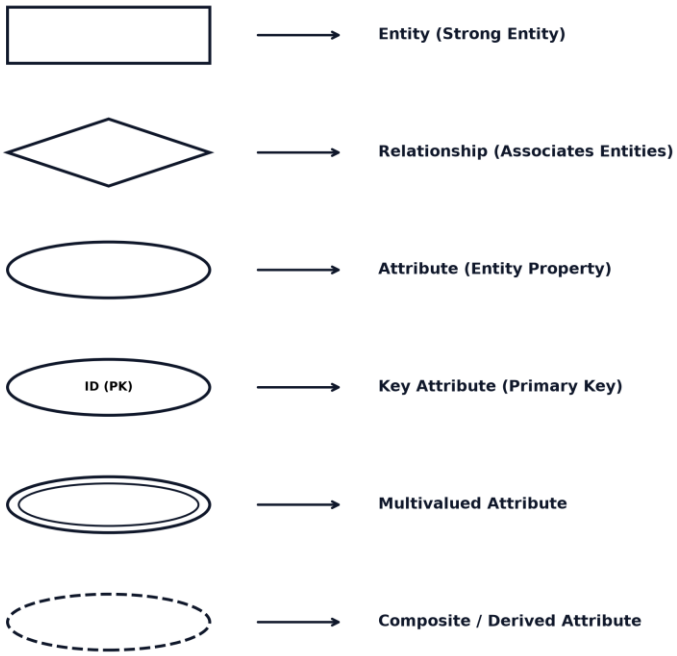


Fig. 8.5.1: ER Diagram Symbols & Conventions

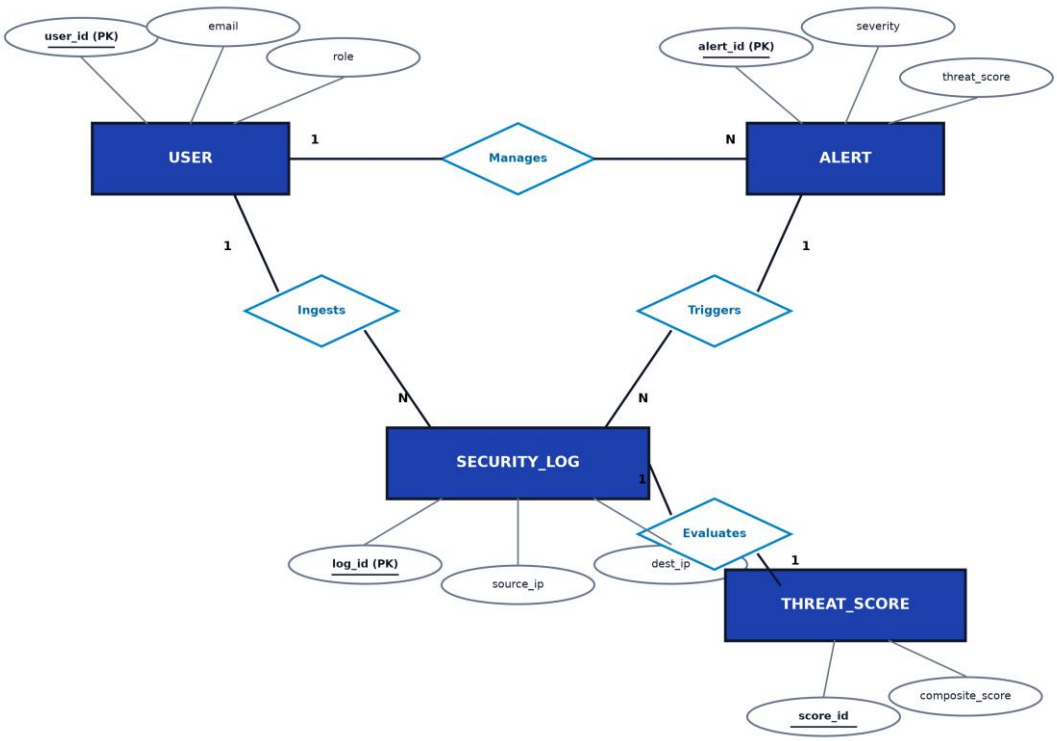


Fig. 8.5.2: Entity-Relationship (ER) Diagram of CyberShield-AI

8.6 DATA DICTIONARY

Table 8.6.1: Data Dictionary - User Entity (PostgreSQL)

Field Name	Data Type	Constraints	Description
id	UUID	Primary Key, Auto-gen	Unique system identifier for analyst/admin account.
email	VarChar(255)	Unique, Not Null	Corporate email address used for login authentication.
password	VarChar(255)	Not Null	Bcrypt 12-round salted password hash.
firstName	VarChar(100)	Not Null	First name of registered user.
lastName	VarChar(100)	Not Null	Last name of registered user.
role	Enum	VIEWER, ANALYST, ADMIN	Role-based access permission level (Default: VIEWER).
isActive	Boolean	Default True	Account status flag indicating if login is permitted.
createdAt	DateTime	Default Now	Timestamp of account registration.
updatedAt	DateTime	Auto Update	Timestamp of last profile update.

Table 8.6.2: Data Dictionary - Security Log Document (MongoDB)

Field Name	Data Type	Constraints	Description
_id	ObjectId	Primary Key, Auto-gen	Unique BSON document identifier in MongoDB.
timestamp	DateTime	Not Null, Indexed	UTC timestamp when network event was generated.
sourceIp	String	Not Null, Indexed	Origin IPv4/IPv6 address of the network connection.
destinationIp	String	Not Null	Target host IPv4/IPv6 address within the network.
sourcePort	Integer	Range 0 to 65535	Origin transport-layer port number.
destinationPort	Integer	Range 0 to 65535	Target service port number (e.g. 22, 80, 443).
protocol	Enum	TCP, UDP, ICMP, DNS...	Network transport protocol employed.
action	Enum	ALLOW, DENY, DROP, ALERT	Action taken by boundary security device.

Field Name	Data Type	Constraints	Description
severity	Enum	LOW, MEDIUM, HIGH, CRITICAL	Severity classification of the raw log record.
isProcessed	Boolean	Default False, Indexed	Flag indicating whether ML threat scoring has executed.

Table 8.6.3: Data Dictionary - Incident Alert Entity (PostgreSQL)

Field Name	Data Type	Constraints	Description
id	UUID	Primary Key, Auto-gen	Unique incident ticket identifier in PostgreSQL.
type	Enum	BRUTE_FORCE, DDOS, MALWARE...	Categorized security incident type.
severity	Enum	LOW, MEDIUM, HIGH, CRITICAL	Incident urgency level determined by ThreatScore.
status	Enum	NEW, INVESTIGATING, RESOLVED...	Current lifecycle state of the incident ticket.
title	VarChar(255)	Not Null	Concise summary title of the security incident.
sourceIp	VarChar(45)	Not Null, Indexed	Malicious or anomalous origin IP address.
threatScore	Float	Range 0.0 to 100.0	Composite risk score computed by ML ensemble.
reasons	JsonB	Not Null	Array of SHAP feature attribution strings.
assignedToId	UUID	Foreign Key -> User.id	Reference to assigned investigating SOC analyst.
createdAt	DateTime	Default Now, Indexed	Timestamp when incident alert was triggered.

Table 8.6.4: Data Dictionary - Threat Score Entity (PostgreSQL)

Field Name	Data Type	Constraints	Description
id	UUID	Primary Key, Auto-gen	Unique risk score entity reference number.
entityType	Enum	IP, USER, SESSION	Type of entity evaluated for security risk.

Field Name	Data Type	Constraints	Description
entityValue	VarChar(255)	Not Null, Indexed	Identifier value of the entity (e.g. IP address or username).
score	Float	Range 0.0 to 100.0	Composite normalized threat risk calculation.
category	Enum	SAFE, LOW, MEDIUM, HIGH, CRITICAL	Operational risk tier assigned to the entity.
factors	JsonB	Not Null	Breakdown of individual anomaly and heuristic weights.
confidence	Float	Range 0.0 to 1.0	Statistical confidence probability of the ML model.
createdAt	DateTime	Default Now, Indexed	Timestamp when threat score was recorded.

CHAPTER 9
LIMITATION AND FUTURE ENHANCEMENTS

- **LIMITATIONS**
- **FUTURE ENHANCEMENTS**

9.1 LIMITATIONS

• **1. Unlabeled Zero-Day Validation:** Because zero-day threats by definition lack historical ground truth, initial anomaly confidence scoring relies on unsupervised decision margins rather than labeled verification.

• **2. Network Ingestion Bandwidth:** Extreme peak bursts exceeding 50,000 logs/second require multi-node Redis clustering to avoid job queue memory saturation on resource-constrained hosts.

• **3. Encrypted Payload Inspection:** Deep payload analysis is restricted when network traffic utilizes end-to-end TLS 1.3 encryption, shifting detection weight to flow telemetry and port metadata.

9.2 FUTURE ENHANCEMENTS

• **1. Automated SOAR Playbook Execution:** Integrate automated Security Orchestration, Automation, and Response (SOAR) playbooks to dynamically reconfigure firewall ACLs and isolate infected endpoints.

• **2. Large Language Model (LLM) Incident Summarization:** Deploy local privacy-preserving LLMs (e.g., Llama 3) to generate natural language executive briefing summaries from multi-source incident logs.

• **3. Graph Neural Network (GNN) Attack Graphing:** Incorporate Graph Neural Networks to model lateral movement across enterprise Active Directory topologies and cloud identity graphs.

• **4. Decentralized Threat Intelligence Sharing:** Implement STIX/TAXII 2.1 protocol standards for federated, privacy-preserving threat intelligence exchange across partner organizations.

CHAPTER 10
CONCLUSION

- **CONCLUSION**

10.1 CONCLUSION

The CyberShield-AI Cyber Threat Intelligence, Detection, Analysis and Attack Prediction Platform demonstrates how modern cloud-native engineering, polyglot persistence, unsupervised machine learning ensembles, and explainable artificial intelligence can transform enterprise cybersecurity. Throughout the project lifecycle, the following milestones were achieved:

- **1. End-to-End Intelligence Ecosystem:** Successfully designed and deployed a responsive Next.js 16 and NestJS 11 platform connecting frontline SOC analysts, threat hunters, and administrators.
- **2. Unsupervised ML Anomaly Detection:** Engineered a high-performance Python FastAPI microservice combining Isolation Forest and One-Class SVM to detect novel cyber anomalies with sub-100ms latency.
- **3. Explainable AI Attribution:** Demystified machine learning predictions by pairing SHAP feature importance attributions with heuristic reasoning, eliminating black-box distrust.
- **4. Predictive Kill-Chain Forecasting:** Delivered multi-stage attack forecasting mapped to MITRE ATT&CK tactics, allowing proactive containment before damage occurs.
- **5. Rigorous Systematic Validation:** Validated all functional, security, and machine learning modules through extensive black-box, white-box, and attack simulation test suites.

In conclusion, CyberShield-AI bridges the critical gap between overwhelming raw security data and actionable defensive intelligence, establishing a robust, scalable, and explainable foundation for next-generation cyber defense operations.

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